NLU Course Project - Part 1: LM (LAB 4)

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1. Introduction

This report presents my contributions and results for Part 1 of the NLU course project at the University of Trento, focused on language modeling for next-word prediction. The work is divided into two sections: Part A and Part B.

- **Part A** incrementally improves the baseline Recurrent Neural Network (RNN) language model by replacing the RNN with a Long Short-Term Memory (LSTM) network, introducing dropout layers, and switching from Stochastic Gradient Descent (SGD) to the AdamW optimizer.
- **Part B** builds upon the best-performing vanilla LSTM model from Part A and explores three advanced regularization techniques: Weight Tying, Variational Dropout, and Non-Monotonically Triggered Averaged SGD (NT-AvSGD).

Across both parts, key hyperparameters, such as learning rate, hidden size, embedding dimension, and batch size, were tuned to minimize perplexity (PPL).

2. Implementation Details

All modifications were applied incrementally, with performance evaluated at each step through loss curves and CSV-based logging. Early stopping with a patience of 3 epochs was employed for all steps, except for when using NT-AvSGD. A test mode was also implemented to re-evaluate saved models and verify the logged results.

In **Part A**, the baseline RNN was improved by:

1. Replacing it with an LSTM [1].
2. Adding two dropout layers [2], one after the embedding layer and one before the final output layer.
3. Switching the optimizer from SGD to AdamW [3].

In **Part B**, starting from the best-performing vanilla LSTM from part A, the following regularization techniques were implemented, as proposed by Merity et al. [4]:

- **Weight Tying**: Weights are shared between the embedding and output layers, reducing parameter count and enforcing consistency. A projection layer was added when dimensions differed, in order to bypass the requirement of matching sizes.
- **Variational Dropout**: The same dropout mask is used across all time steps, maintaining temporal consistency and improving generalization. The implementation was made by following the work proposed in [5].
- **NT-AvSGD**: An enhanced SGD variant where averaging is triggered only after the validation perplexity fails to improve over a specified number of evaluations. As described in the reference paper, the switch is performed when the validation loss exceeds the minimum of a past loss window, defined by a non-monotonic interval. Once triggered, model weights are averaged until the end of training. This approach stabilizes training and enhances generalization. The method was implemented by adapting the paper's pseudocode into a Pythonic implementation.

3. Results

Extensive hyperparameter tuning was performed to optimize model performance. Perplexity (PPL) was used as the evaluation metric, with a target threshold of 250 to surpass. The final results and corresponding hyperparameter values for the best-performing models are presented in the tables on the last page. A summary of key findings and conclusions for each part is provided below.

Part A: As shown in Table 1, all proposed modifications improved performance, with all configurations achieving a PPL below 250 when properly tuned. Key observations include:

- **Model Size and Batch Size**: A balanced embedding and hidden sizes (200/300) prevented overfitting, which was observed with larger models. Smaller batch sizes (32, 64) resulted in more stable training than larger ones (128), with 64 providing the best results overall.
- **LSTM with SGD**: Using LSTM with SGD, very low learning rates (0.0001–0.01) caused underfitting and slow convergence, while rates higher than 1.0 improved convergence but risked overfitting if too high. The best PPL (141.07) was achieved at a learning rate of 2.0.
- **Dropout**: Adding dropout improved generalization, lowering PPL to 123.77. A dropout probability of 0.3 gave slightly better results than other tested values (0.1–0.5).
- **AdamW Optimizer**: Switching to AdamW further boosted performance (PPL = 114.69), especially with moderate learning rates (5×10^{-4}), highlighting the benefits of this optimizer with respect to vanilla SGD.

Part B: Building on Part A's best vanilla LSTM model (with SGD), additional regularization and optimization techniques were explored, confirming and extending previous findings. Results are shown in Table 2.

- **Weight Tying**: Improved baseline performance, lowering PPL to 121.99.
- **Variational Dropout**: Further mitigated overfitting and improved generalization, reducing PPL to 113.17.
- **NT-AvSGD Optimizer**: Recommended settings were used, included a logging interval equal to the number of iterations per epoch and a non-monotonic interval of 5. Combining Weight Tying, Variational Dropout, and NT-AvSGD leverages both structural regularization and training stabilization, yielding the best generalization (PPL = 95.53).

4. References

[1] "torch.nn.lstm," https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html, accessed: 2025-05-26.

[2] "torch.nn.dropout," https://pytorch.org/docs/stable/generated/torch.nn.Dropout.html, accessed: 2025-05-26.

[3] "torch.optim.adamw," https://pytorch.org/docs/stable/generated/torch.optim.AdamW.html, accessed: 2025-05-26.

[4] S. Merity, N. S. Keskar, and R. Socher, "Regularizing and optimizing lstm language models," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2018. [Online]. Available: https://arxiv.org/abs/1708.02182

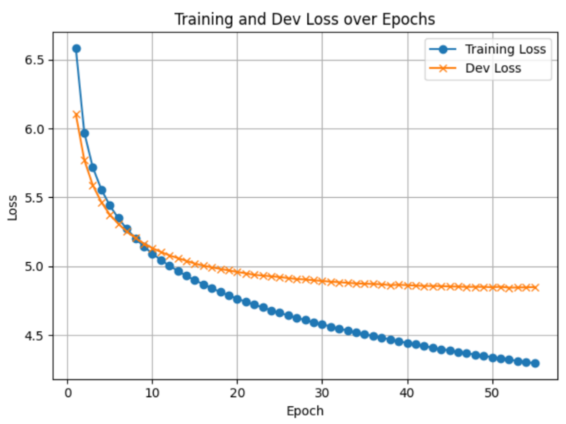
[5] S. Merity, "locked_dropout.py - awd-lstm language model," https://github.com/salesforce/awd-lstm-lm/blob/master/locked$_dropout.py$.

Table 1: *Part A: Incremental Model Improvements with Hyperparameters and test results of the best saved models. The best result is highlighted.*

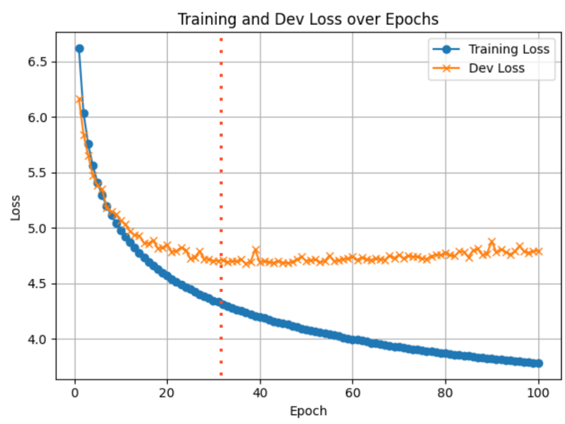
Model Config	lr	Batch Size	Hid/Emb Size	Out/Emb Dropout	Test PPL
Basic RNN	1.0	64	200/300	–	161.33
Vanilla LSTM	2.0	64	200/300	–	141.07
+ Dropout	2.0	64	200/300	0.3	123.78
+ AdamW	0.0005	64	200/300	0.3	**114.69**

Table 2: *Part B: Regularization Techniques with Hyperparameters and test results of the best saved models. The best result is highlighted.*

Model Config	lr	Batch Size	Hid/Emb Size	Dropout	Test PPL
Vanilla LSTM	2.0	64	200/300	–	141.07
+ Weight Tying	2.0	64	200/300	–	121.99
+ Variational Dropout	2.0	64	200/300	0.3	113.17
+ NT-AvSGD	2.0	64	200/300	0.3	**95.53**



(a) *Training and validation loss curves of the best model from part A.*



(b) *Training and validation loss curves of the best model from part B. A red line indicates where NT-AvSGD averaging started.*

Figure 1: *Loss curves of the best models of each part.*